

2018-Ph-H2-Q2

Section: Our Dynamic Universe
Topic: Forces, Energy and Power

Summary:
A delivery drone (mass 5.50 kg) carries a package (mass 1.25 kg).

- (a)(i) Calculate the upward force produced while hovering.
- (a)(ii) Find the power dissipated by the motor (12 V, 6 Ω).
- (a)(iii) Describe the drone’s motion if the cable breaks.
- (b) Two drones (at 35° cables) support a 3.4 kg package. Find the tension in each cable.

Solution:

(a)(i) Upward force (hovering):

Total mass:
 $m = 5.50 + 1.25 = 6.75 \text{ kg}.$

Weight (upward force to hover):
 $F = mg = 6.75 \times 9.8 = 66 \text{ N}.$

Answer: 66 N.

(a)(ii) Power dissipated by motor:

Current:
 $I = \frac{V}{R} = \frac{12}{6} = 2.0 \text{ A}.$

Power:
 $P = I^2 R = 2.0^2 \times 6 = 24 \text{ W}.$

Answer: 24 W.

(a)(iii) Motion after cable breaks:

- The drone’s upward thrust is momentarily **greater than its weight**, producing upward acceleration.
- Therefore, the drone **accelerates upwards** until thrust is reduced.

Answer: The drone accelerates upwards because the upward force exceeds its weight.

(b) Tension in each cable:

Weight of package:
 $W = mg = 3.4 \times 9.8 = 33.3 \text{ N}.$

Two tensions T share vertical support:
 $2T \cos 35^\circ = 33.3 \Rightarrow T = \frac{33.3}{2 \cos 35^\circ} \approx 20 \text{ N}.$

Answer: 20 N per cable.

Guidance for Students:

- Total weight = combined mass $\times g$.
- For angled cables, use vertical components $T \cos \theta$ to balance weight.
- Hovering means forces are balanced (net force = 0).

Revision Tips:

- **Remember:** $W = mg$.
- **Electrical power:** $P = V^2 / R = I^2 R$.
- Draw force diagrams for angled tensions.

